Practice Problem 1

Ultraviolet light with a wavelength of 300 nm is propagated through a single slit. The 3rd minimum diffraction is at the angle of 45° relative to the horizontal axis.

a) Calculate the width of the slit.

Solution.

$$\lambda = 300\,nm = 300 \times 10^{-9}\,m$$

at $m = 3$; $\theta_3 = 45^\circ$

$$a \sin\theta_m = m\lambda$$

$$a = \frac{m\lambda}{\sin\theta_m} = \frac{3(300 \times 10^{-9}\,m)}{\sin(45^\circ)}$$

$$= 1.273 \times 10^{-6}\,m$$

b) At what angle is the first minimum generated and what is the total number of minimum points on the screen.

Solution.

$$a \sin\theta_m = m\lambda$$

1st minimum $\rightarrow m = 1$

$$(1.273\times10^{-6}\ m)\sin\theta_1 = 300\times10^{-9}\ m$$

$$\theta_1 = \sin^{-1}\left[\frac{300\times10^{-9}}{1.273\times10^{-6}}\right] \approx \underline{13.63^\circ}$$

$\theta_m = 90^\circ \rightarrow \sin\theta_m = 1$ (At max)

$$m = \frac{a}{\lambda} = \frac{1.273\times10^{-6}}{300\times10^{-9}} = 4.243$$

$\therefore$ Total number of minima

$(4.243 \approx 4)(2) = \underline{8}$ points.

(4)

c) If the slit is decreased to 1 μm, how would this new setting influence the diffraction phenomena?

Solution.

1/ Let's determine the total number of minimum points.

$$m = \frac{a}{\lambda} = \frac{1\times10^{-6}\ m}{300\times10^{-9}\ m} = \underline{3.33}$$

⇒ Therefore there is a reduction in the total number of minimum points : $(3.33 \approx 3)(2) = 6$ points.

2/ How about the angle of the 1st minima?

$$\theta_1 = \sin^{-1}\left[\frac{\lambda}{a}\right] = \sin^{-1}\left[\frac{300\times10^{-9}}{1\times10^{-6}}\right]$$

$$= 17.46^\circ \quad (\text{Increase in } \theta)$$